

Lecture 5: Personal Protective Equipment (PPE)



Protection Matters



Objectives

- ▶ Understand different types of PPE and their applications.
- ▶ Learn how to select, maintain, and use PPE correctly.
- ▶ Recognize the limitations of PPE.



What is PPE?

- ▶ **Definition** - Equipment worn to minimize exposure to hazards that cause injuries and illnesses.
- ▶ **Examples** - Gloves, eye protection, hearing protection, hard hats, respirators.



Types of PPE

Type	Application
Head Protection	Helmets, hard hats (construction, falling objects)
Eye & Face Protection	Goggles, face shields (chemical splashes, flying particles)
Hearing Protection	Earplugs, earmuffs (loud environments)
Respiratory Protection	Masks, respirators (airborne particles)
Hand Protection	Gloves (chemicals, cuts, heat)
Foot Protection	Safety boots (heavy objects, punctures)
Body Protection	Coveralls, vests (chemicals, high-visibility)

Types of PPE

Helmets, hard hats (construction, falling objects)



Equipment	Use
1. Safety Helmet 	Safety helmets must be worn when performing most construction works, works in the vicinity of lifting equipment and suspended loads, works in forestry, works in cisterns, wells, shafts, tunnels, etc. In these situations, the use of a safety helmet will help protect against injuries to the scalp, skull and cervical vertebrae.
2. Bump caps 	Bump caps are safety equipment that protect the wearer from minor head bumps and lacerations, but not from falling or flying objects
Specific equipment for operations in the workshop	
3. Hard hat 	Hard hats are used in workplace environments such as industrial or construction sites to protect the head from injury due to falling objects, impact with other objects, debris, rain, and electric shock. Suspension bands inside the helmet spread the helmet's weight and the force of any impact over the top of the head.

Types of PPE

Goggles, face shields
(chemical splashes, flying
particles)



Types of PPE

Earplugs, earmuffs (loud environments)

JOY OF HEARING®
A Complete Hearing & Speech Solution

Safety at Work: Hearing Protection

Use:

- Earmuffs
- Earplugs
- Ear caps



Clinic Timings :
Monday to Saturday: 10 AM to 7 PM
Sundays: Closed



Types of PPE

Masks, respirators
(airborne particles)



Suitable for Many Types of Respirator

Compatible with 6200, 6800, 7502, FF402, 6001cn respirator



Types of PPE

Gloves (chemicals, cuts, heat)



Types of PPE

Safety shoes (heavy objects, punctures)



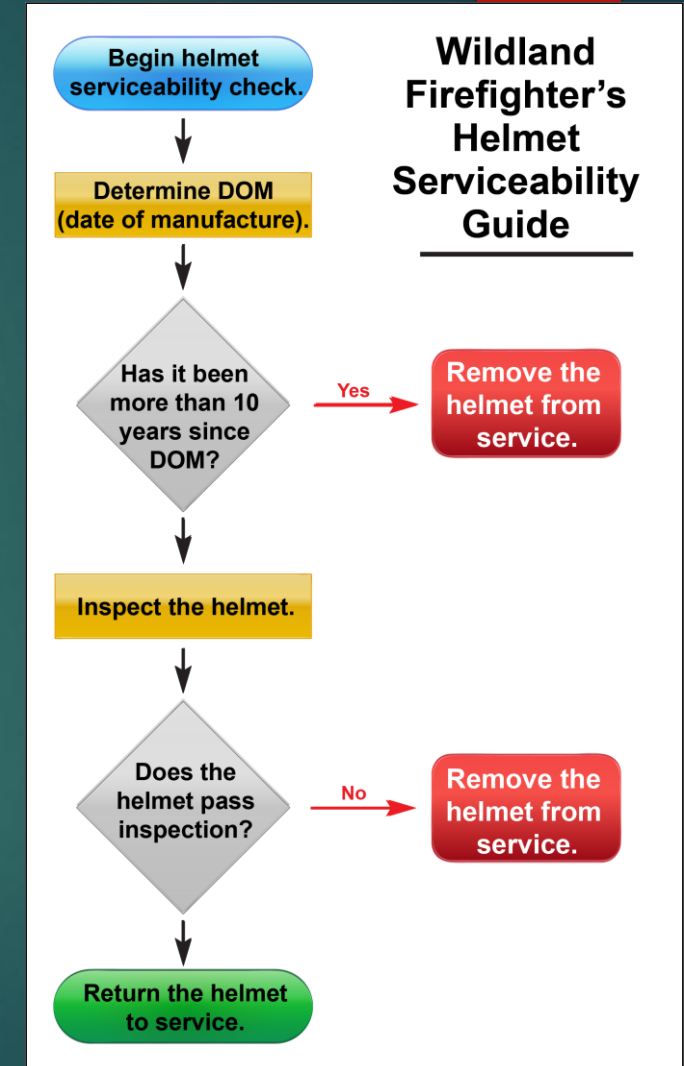
Types of PPE

Coveralls, vests (chemicals, high-visibility)



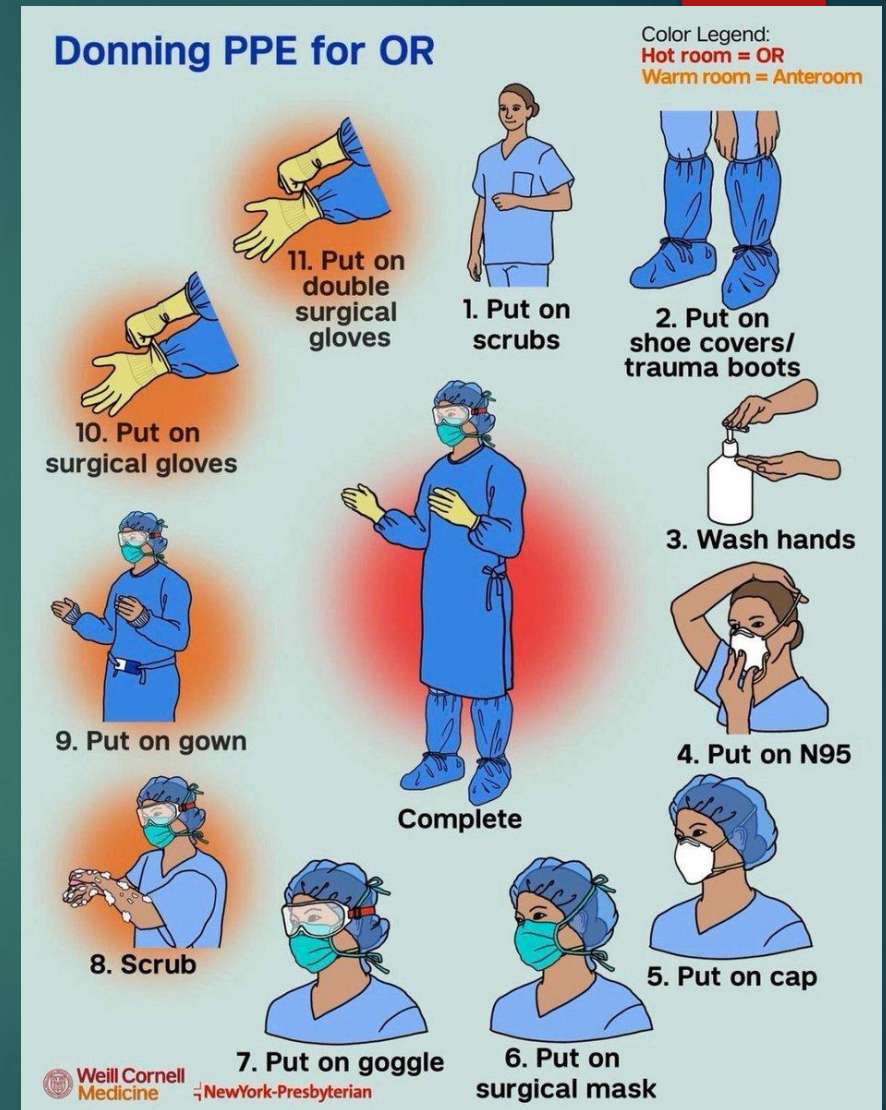
Selecting the Right PPE

- **Assess Hazards** - Identify hazards in the workplace.
- **Match PPE to Risks** - Select PPE that provides appropriate protection.
- **Consider Comfort & Fit** - Ensures compliance and effectiveness.
- **Regulatory Standards** - Follow OSHA, ISO, and local guidelines.
- **Visual** - Flowchart showing PPE selection steps.



Using PPE Correctly

- **Proper Fit** - Ensures full protection.
- **Training** - Workers must be trained to use PPE.
- **Donning & Doffing** - Correct procedures to avoid contamination.
- **Regular Checks** - Inspect before use for damage.



Maintaining PPE

- **Cleaning** - Regularly clean to maintain hygiene.
- **Storage** - Keep in designated, clean areas.
- **Inspection** - Look for wear, cracks, or damage.
- **Replacement** - Follow manufacturer's lifespan guidelines.
- **Visual** - Checklist for PPE maintenance.

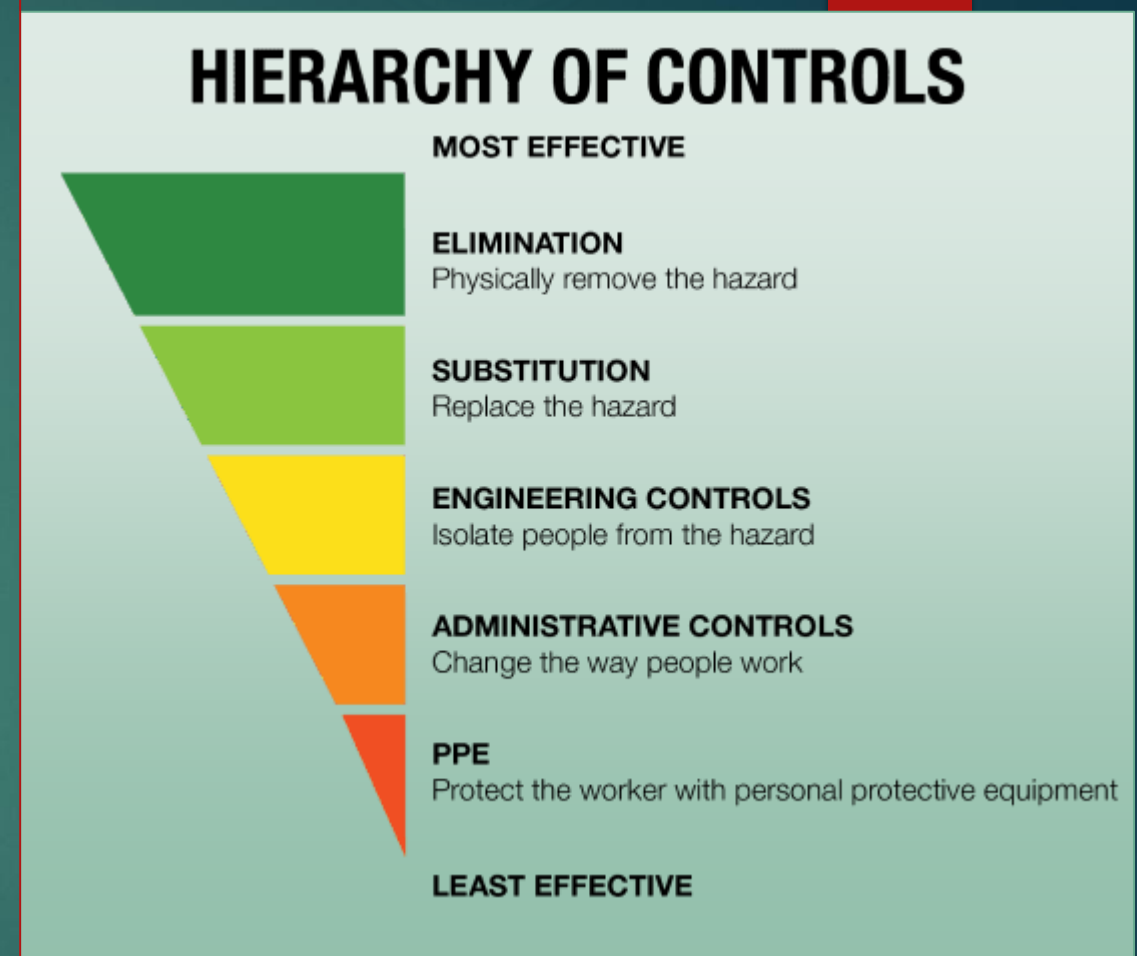
Personal Protective Equipment (PPE) – Checklist

CATEGORY	Y	N	NA	COMMENTS/CORRECTIVE ACTIONS/DATE COMPLETED
Eye and Face Protection - provided for: flying particles, molten material, liquid chemicals, acids, caustic liquids, chemical gases/vapors, and/or injurious light radiation - hazard assessment completed: date ___/___/___ ■ detachable side protectors (flying particles) provided ■ prescription lenses provided or able to cover ■ lens shade in accordance with Table 133 (a)(5) - provided by company or employee - written program in place - documented employee training completed: date ___/___/___ - inspection completed: date ___/___/___ - appropriate fit for employee - equipment maintenance plan/schedule in place OSHA Standard: General Industry 1910.133 / Construction 1926.102 ANSI Standard: Z87.1-2010 / Z87.1-2003 / Z87.1-1989				
Respiratory Protection - provided for: dust, fog, fumes, mists, gases, smoke, sprays, and/or vapors (when engineering controls are not feasible) - hazard assessment completed: date ___/___/___ - provided by company or employee - written program in place - documented employee training completed: date ___/___/___ - inspection completed: date ___/___/___ - medical evaluation completed: date ___/___/___ - fit test completed: date ___/___/___ - equipment maintenance plan/schedule in place OSHA Standard: General Industry 1910.134 (App A, B-1, B2, C, & D) / Construction 1926.103				
Head Protection - provided for: falling objects and/or electrical shock - hazard assessment completed: date ___/___/___ - provided by company or employee - written program in place - documented employee training completed: date ___/___/___ - inspection completed: date ___/___/___ - appropriate fit for employee - equipment maintenance plan/schedule in place OSHA Standard: General Industry 1910.135 / Construction 1926.102 ANSI Standard: Z89.1-2003 / Z89.1-1997				

continued...

Limitations of PPE

- **Not a Substitute** - PPE does not eliminate hazards.
- **Human Error** - Improper use can reduce effectiveness.
- **Comfort** - Can cause discomfort, leading to non-compliance.
- **Limited Protection** - Each PPE type has specific protection limits.



PERSONAL PROTECTIVE EQUIPMENTS

1



Clothing

2



Respiratory

3



Eye & Face

4



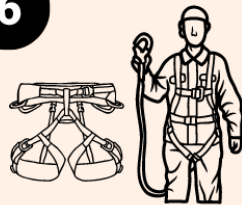
Head

5



Hearing

6



Harnesses

7



Skin

8



Footware

9



Sun & Heat

10

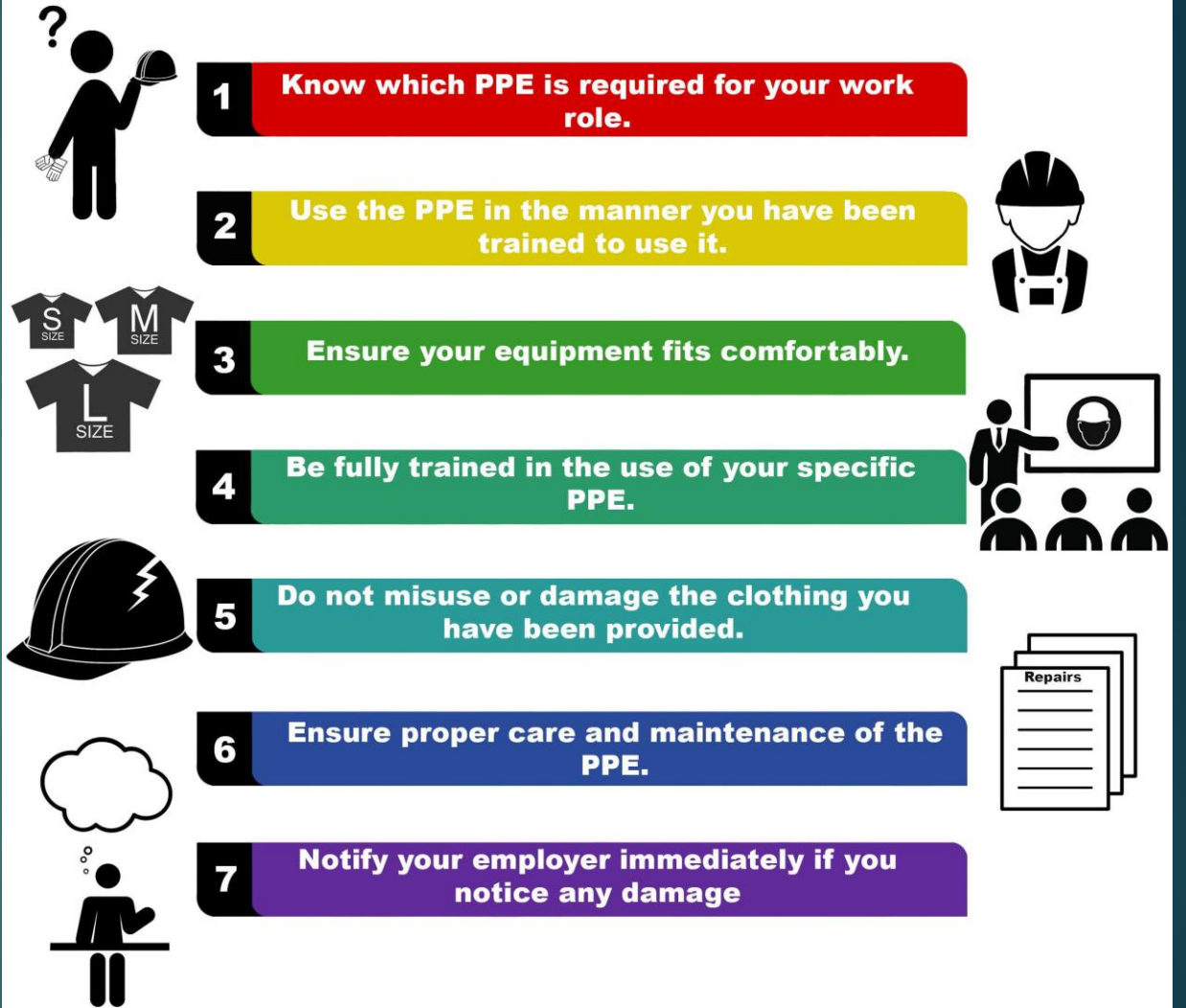


Disposable
Clothing

Top 10 items used in Personal Protective Equipment (PPE)



Rules for using Personal Protective Equipment (PPE)



Summary

- PPE protects workers from injuries and illnesses.
- Selection, use, and maintenance are critical for effectiveness.
- PPE has limitations; always complement it with other controls.

